

Copyright Project Presentation

On

EMOGALAXY – EMOTION FACE DETECTION

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Existing emotion detection systems rely on single input sources such as facial expressions, leading to inaccurate results.

They lack real-time processing, behavioral insights, and multi-person support.

Most systems do not integrate voice-based emotion analysis or predictive capabilities.

There is a need for a unified system that can analyze emotions using multiple modalities, provide real-time insights, and predict emotional patterns.

EmoGalaxy addresses these challenges using multimodal AI techniques, behavioral analysis, and emotion prediction.

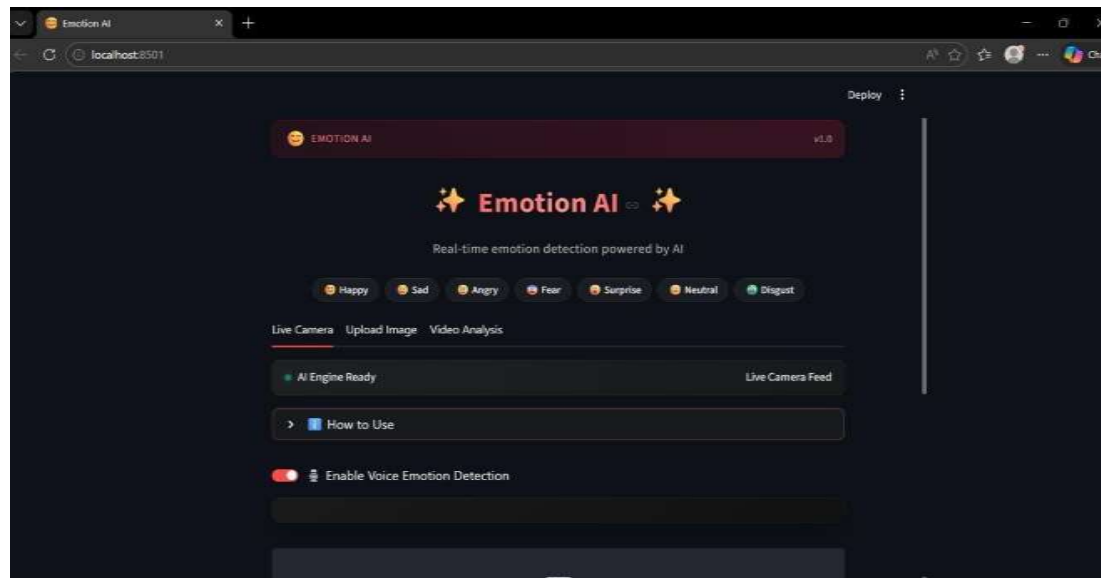
PROJECT OVERVIEW

EmoGalaxy is an AI-based emotion detection system designed to analyze human emotions using facial expressions, voice signals, and behavioral patterns.

The system supports real-time webcam detection, image-based analysis, and video-based emotion tracking.

It integrates multiple modules including face detection, voice emotion analysis, multimodal fusion, behavioral analytics, and emotion prediction.

The platform provides interactive visualizations, real-time insights, and downloadable reports, making it a comprehensive solution for emotion-aware applications.



Frontend

- Streamlit
- Plotly
- Matplotlib

Backend / Processing

- Python
- OpenCV
- DeepFace
- Librosa
- NumPy, Pandas

- **Real-Time Emotion Detection**

Detects human emotions instantly using live webcam input with high accuracy.

- **Voice Emotion Analysis** ☐

Analyzes speech signals using audio features like pitch and MFCC to detect emotions.

- **Multimodal Emotion Fusion**

Combines facial and voice emotions to generate a more reliable final prediction.

- **Image & Video Emotion Analysis**

Supports emotion detection from uploaded images and videos with detailed insights.

KEY FEATURES

- **Behavioral Analysis**

Calculates stability score, volatility index, and detects stress patterns over time.

- **Multi-Person Tracking**

Tracks multiple individuals simultaneously and assigns unique IDs for separate analysis.

- **Emotion Prediction**

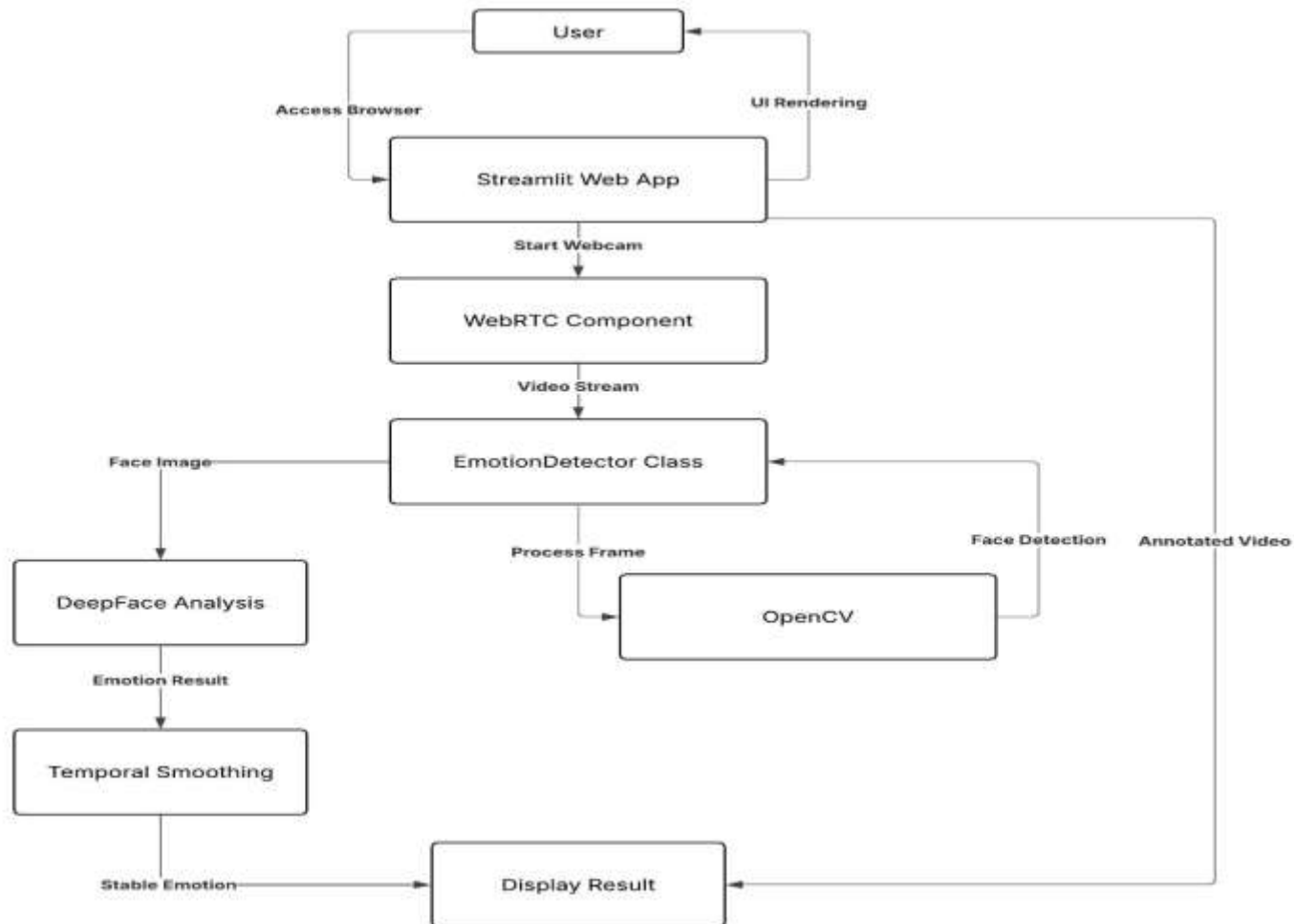


Predicts future emotional states using a Markov-based model.

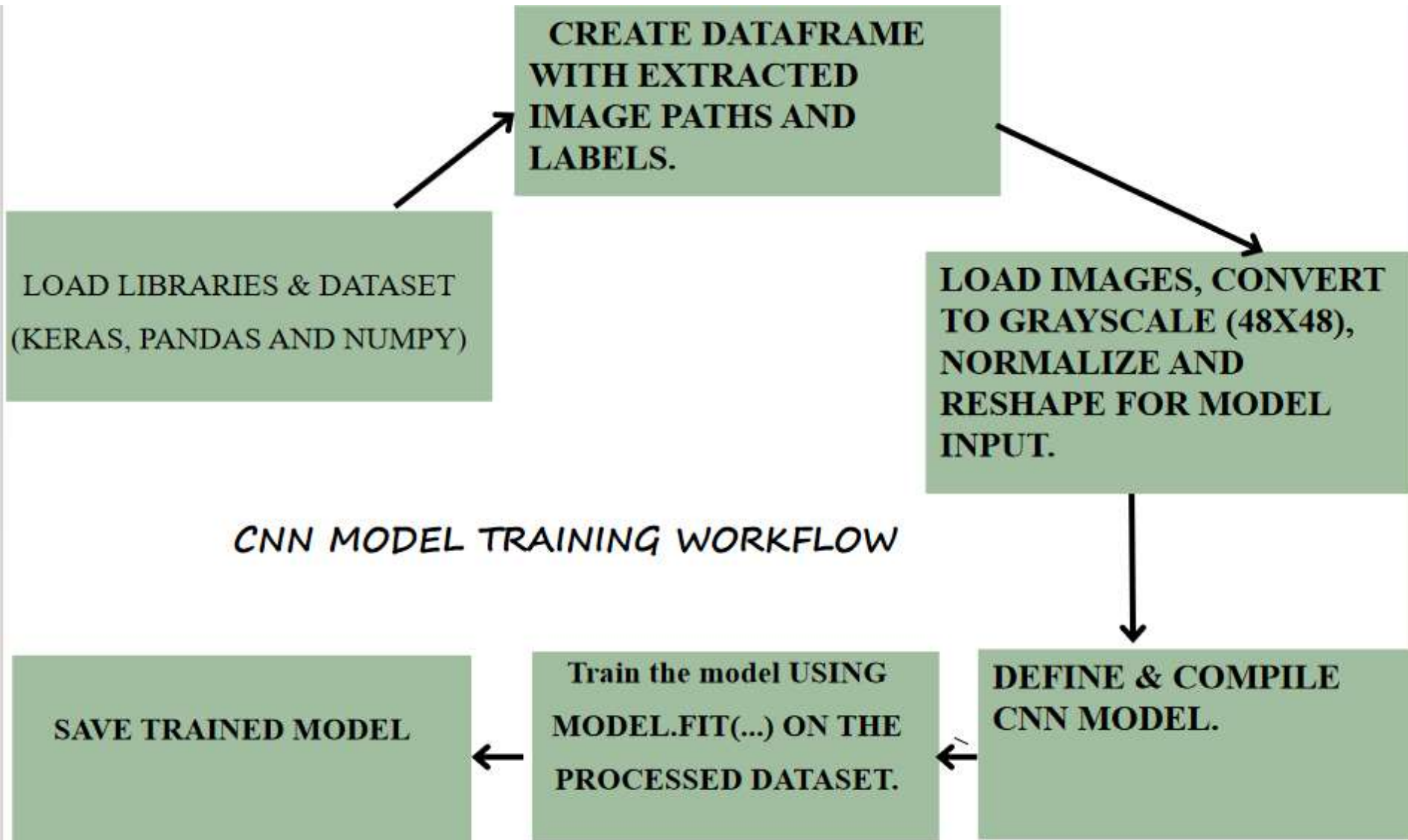
- **Interactive Visualizations & Reports**

Provides charts, timelines, and downloadable reports for better insights.

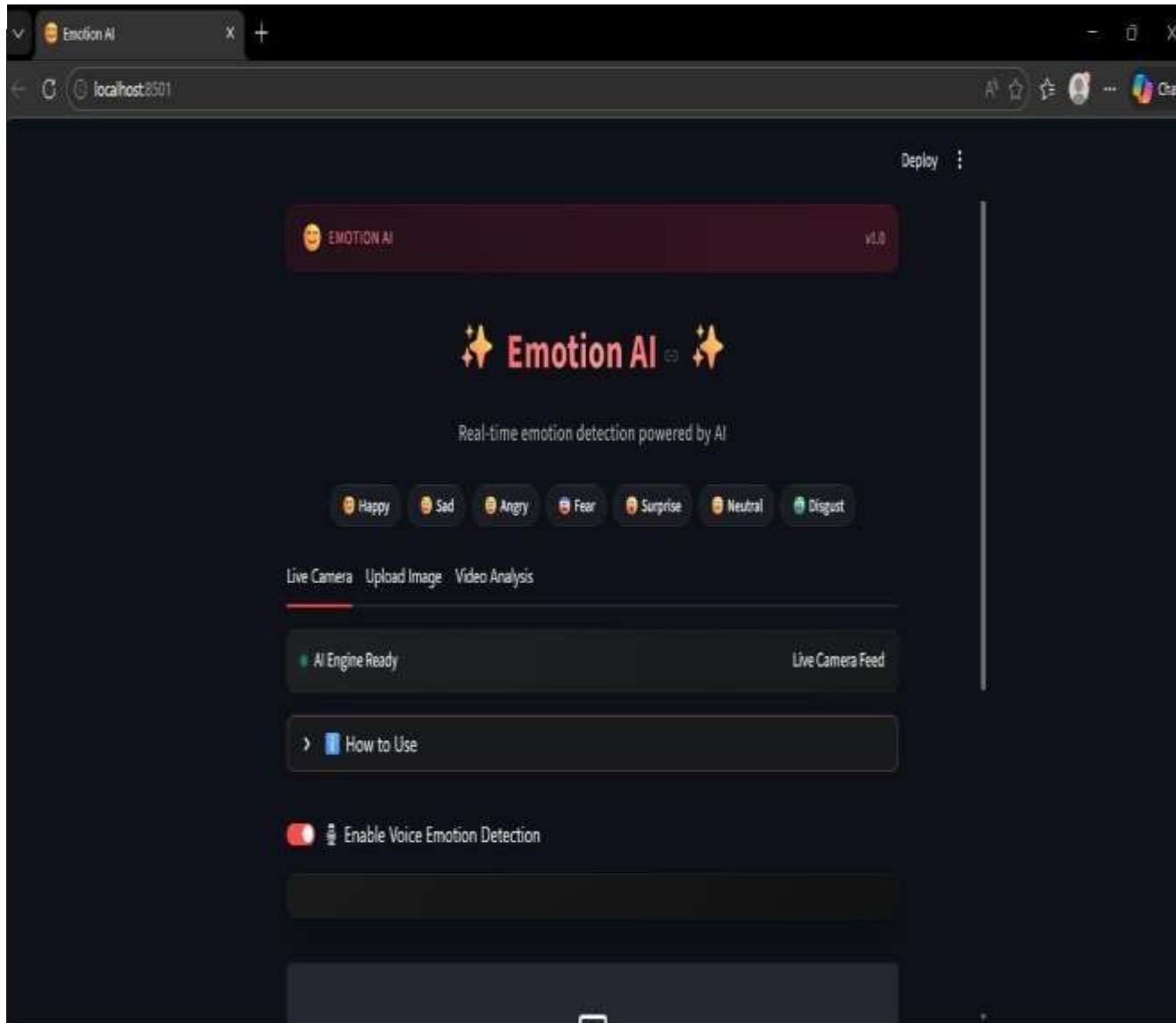
FLOWCHART



FLOWCHART

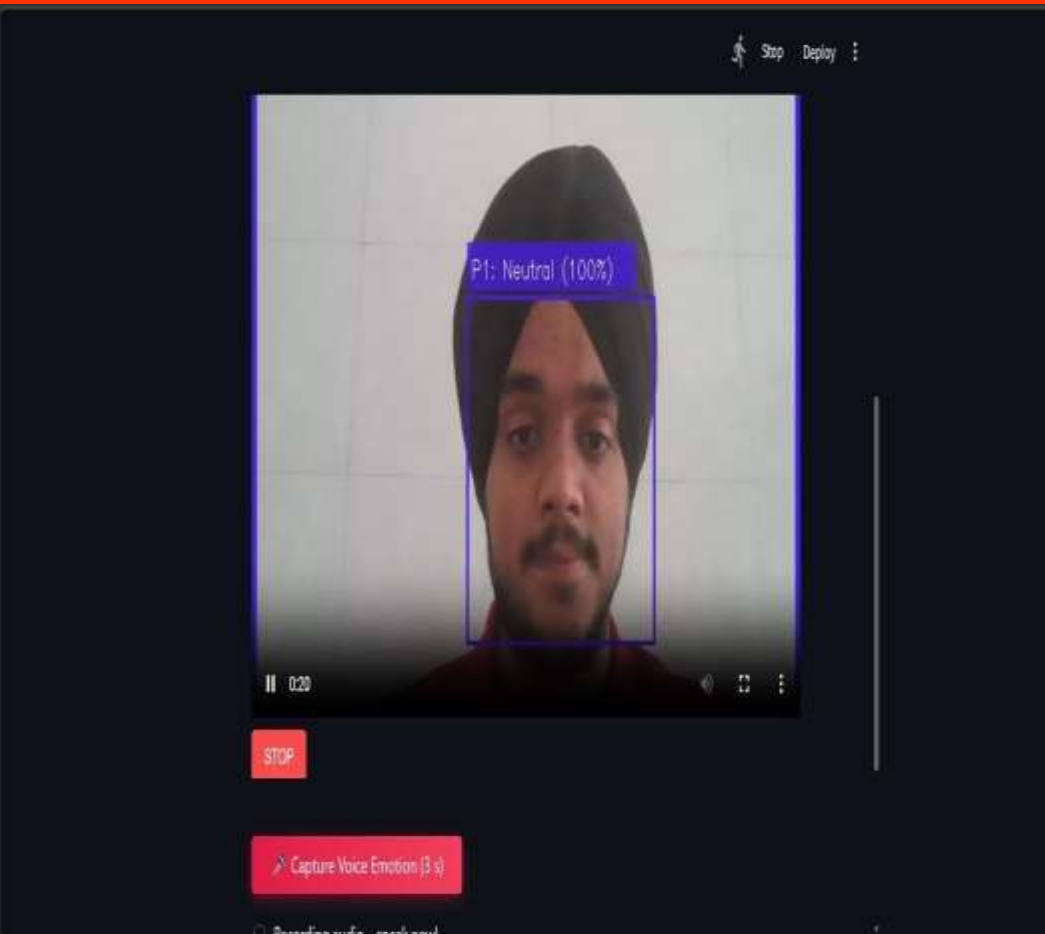


RESULTS



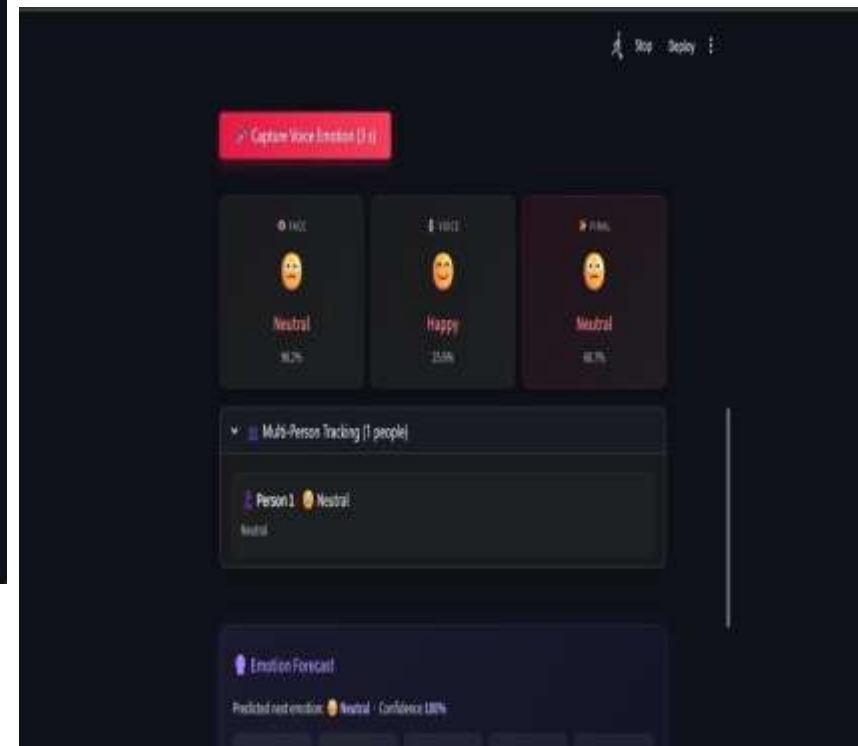
Home Page of EmoGalaxy:
Clean and modern UI with
tab-based navigation for
Live Camera, Image
Upload, and Video
Analysis.

RESULTS

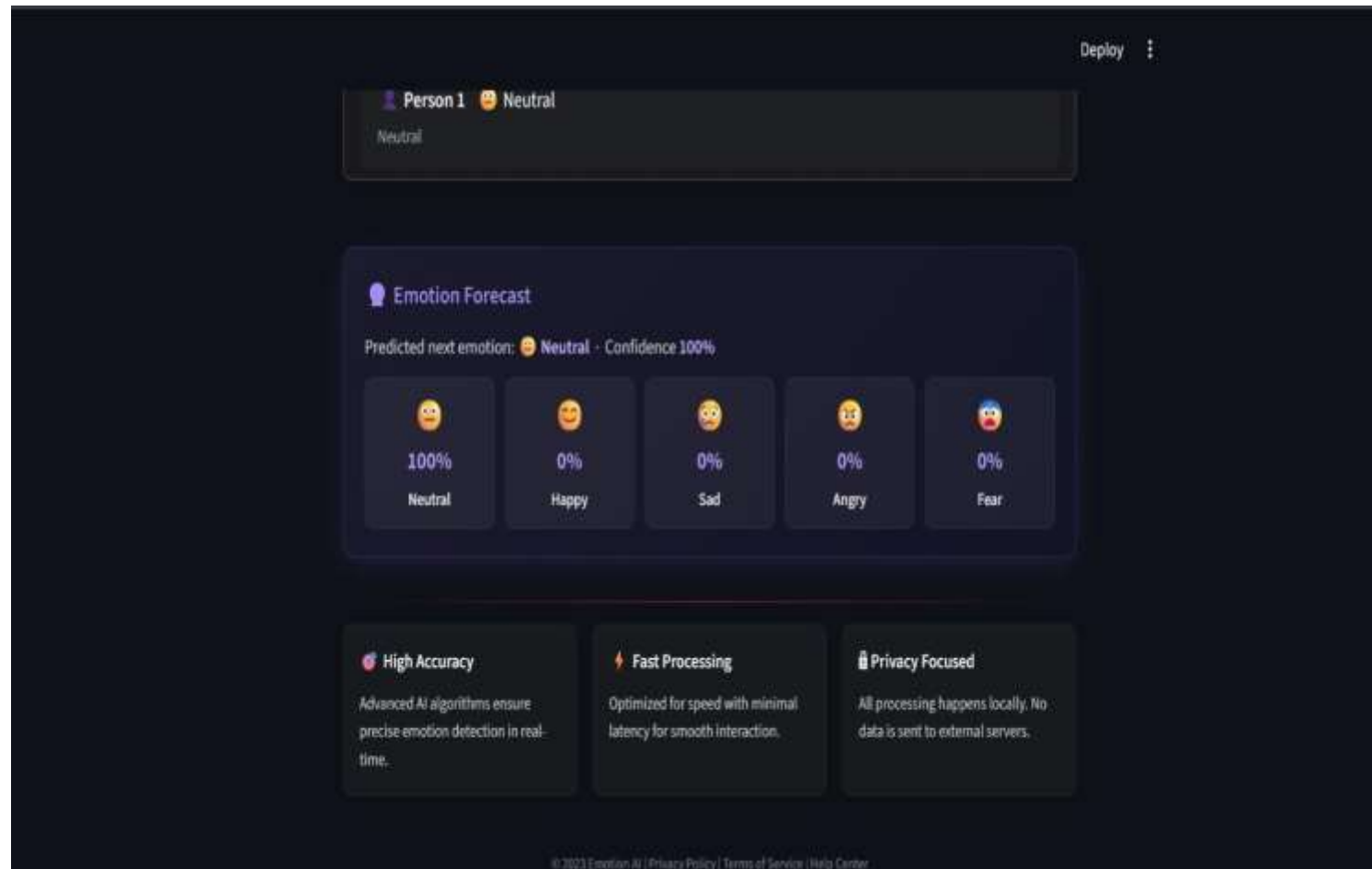


Live Camera Module:

- Real-time face detection
- Emotion labels + confidence
- Voice emotion detection
- Multimodal output



RESULTS

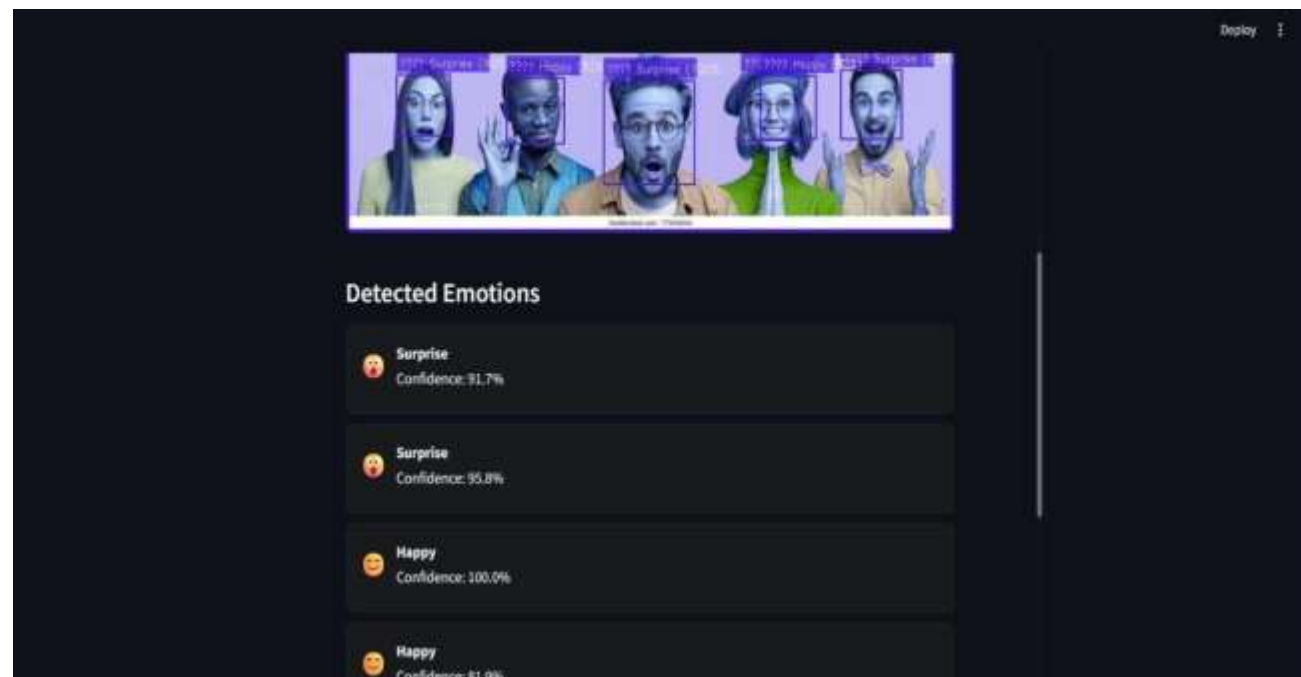


RESULTS

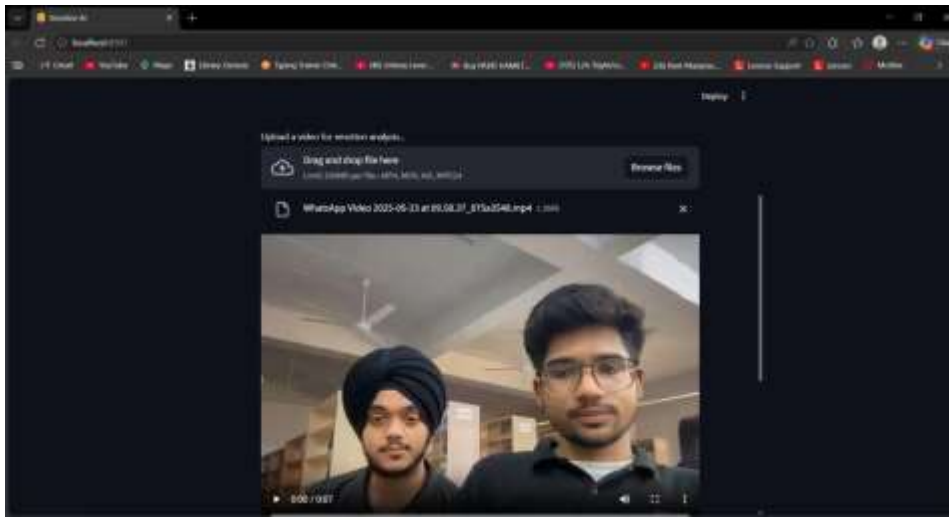


Upload Module:

- Upload image
- Detect multiple faces
- Emotion + confidence

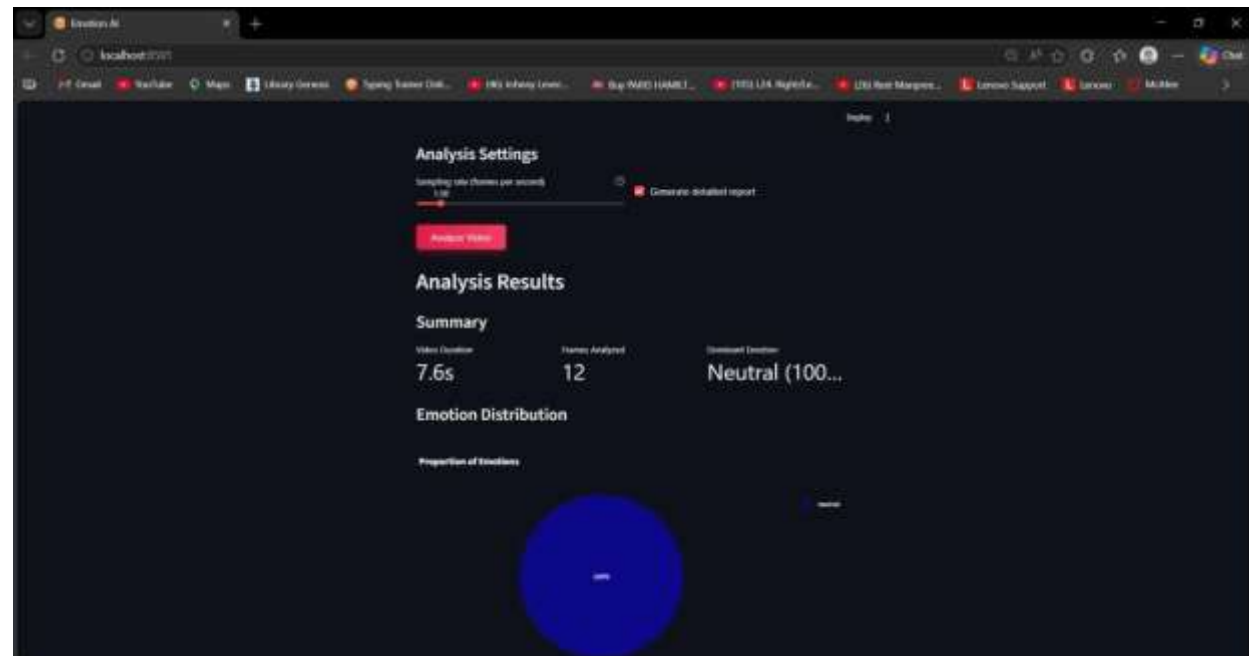


RESULTS

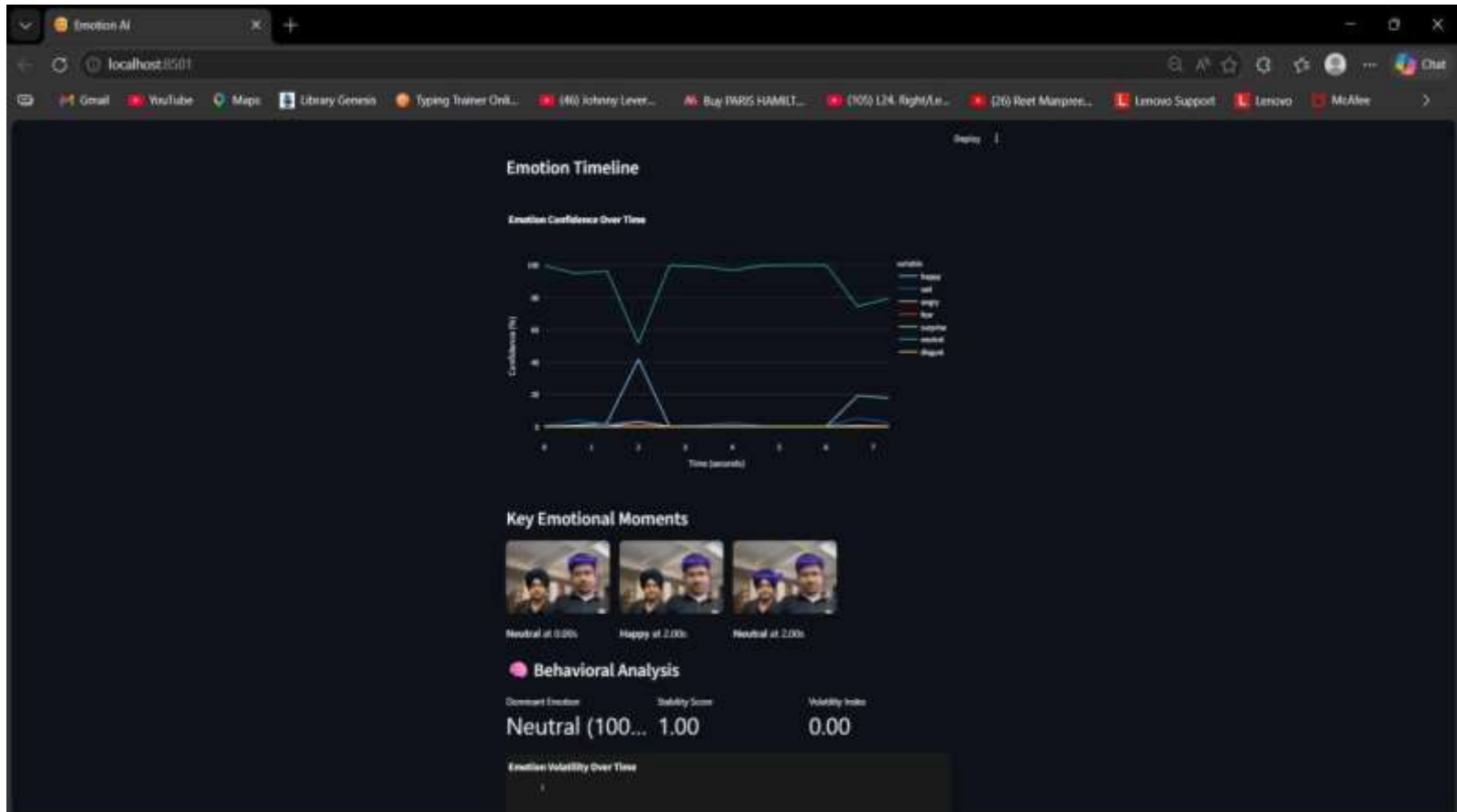


Video Analysis Module:

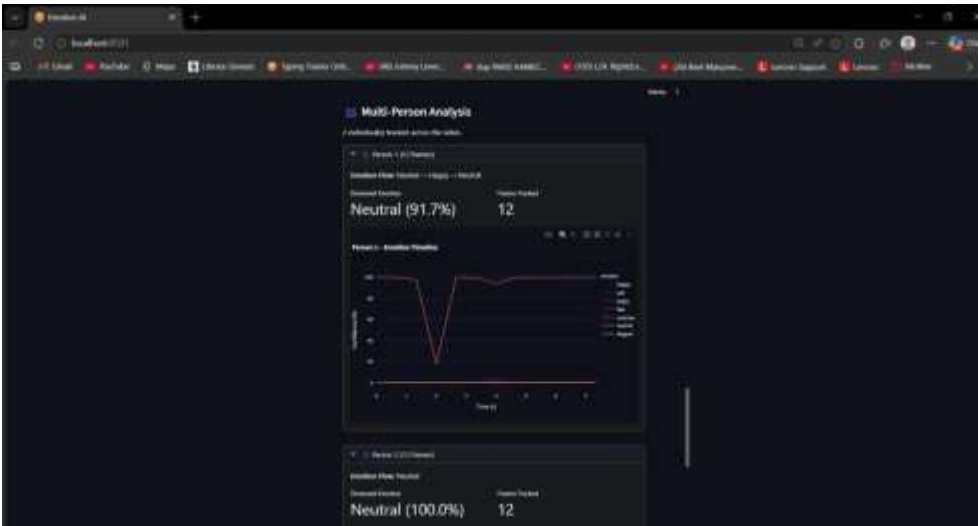
- Timeline graphs
- Emotion distribution
- Key moments



RESULTS

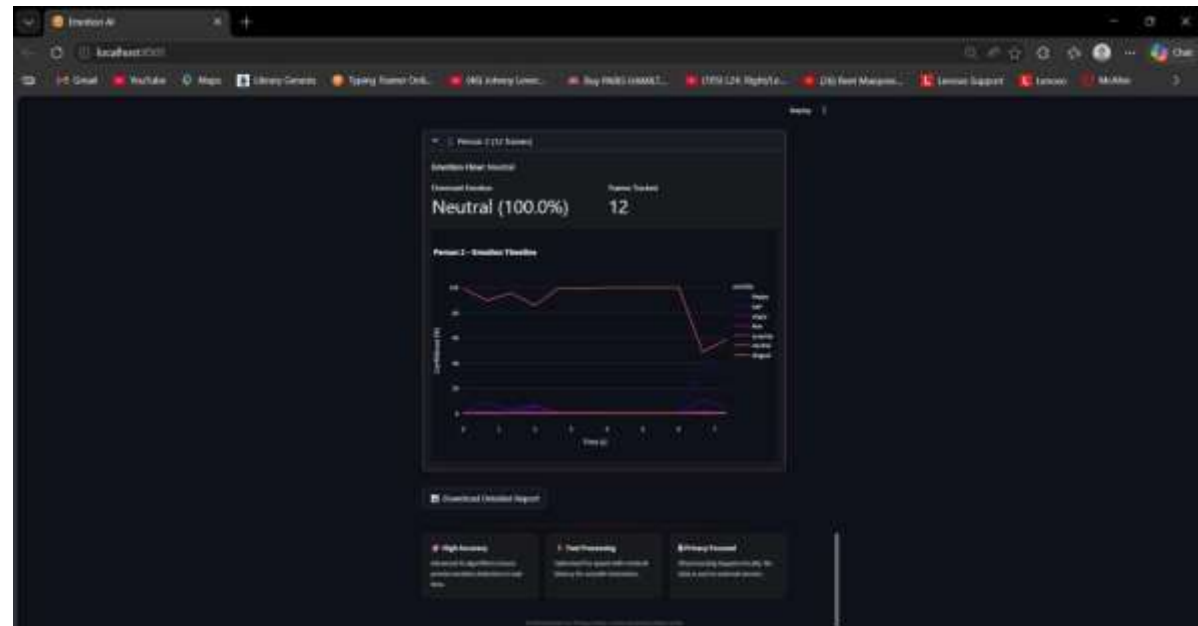


RESULTS

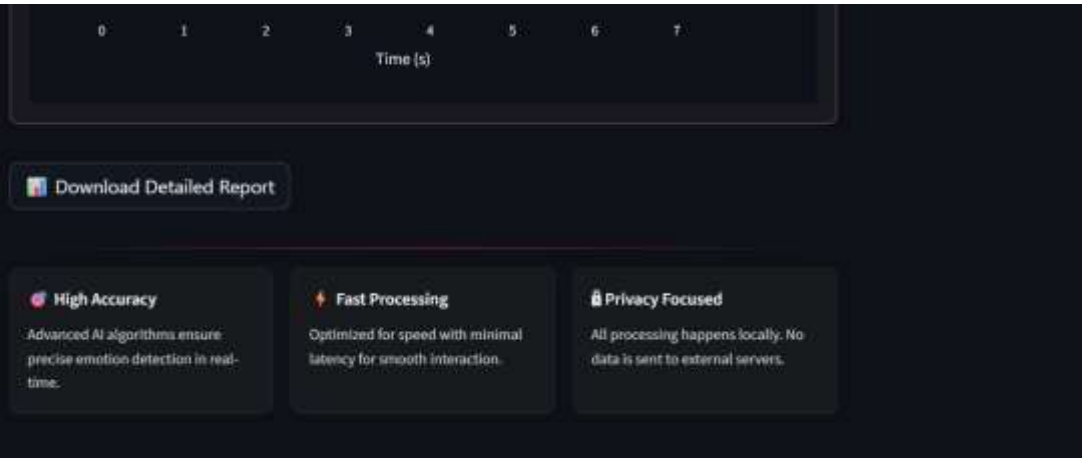


Behavioral Analysis:

- Stability Score
- Volatility Index
- Stress detection

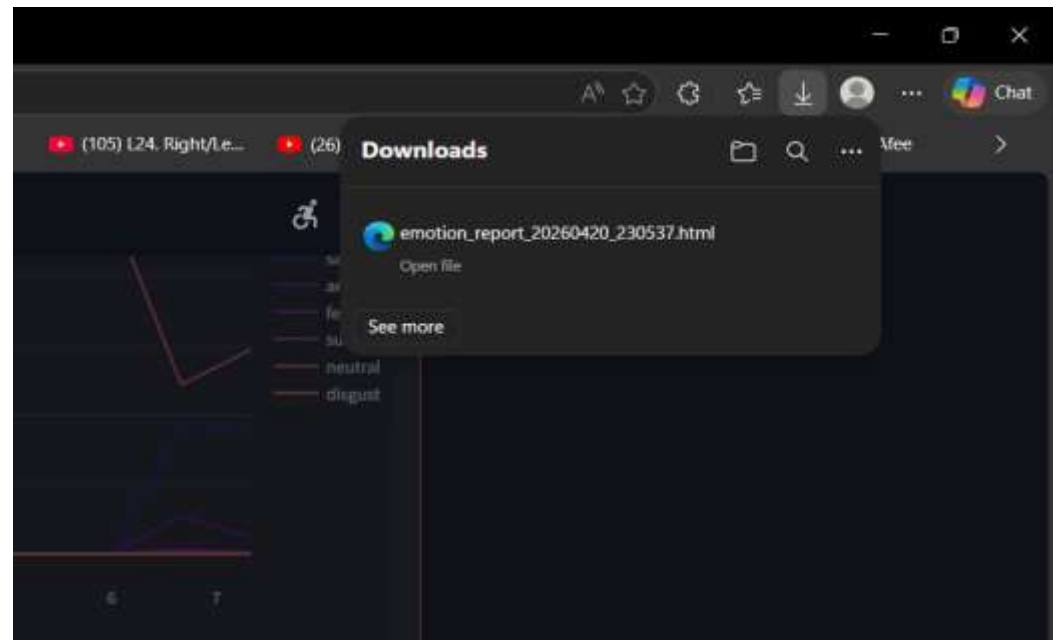


RESULTS

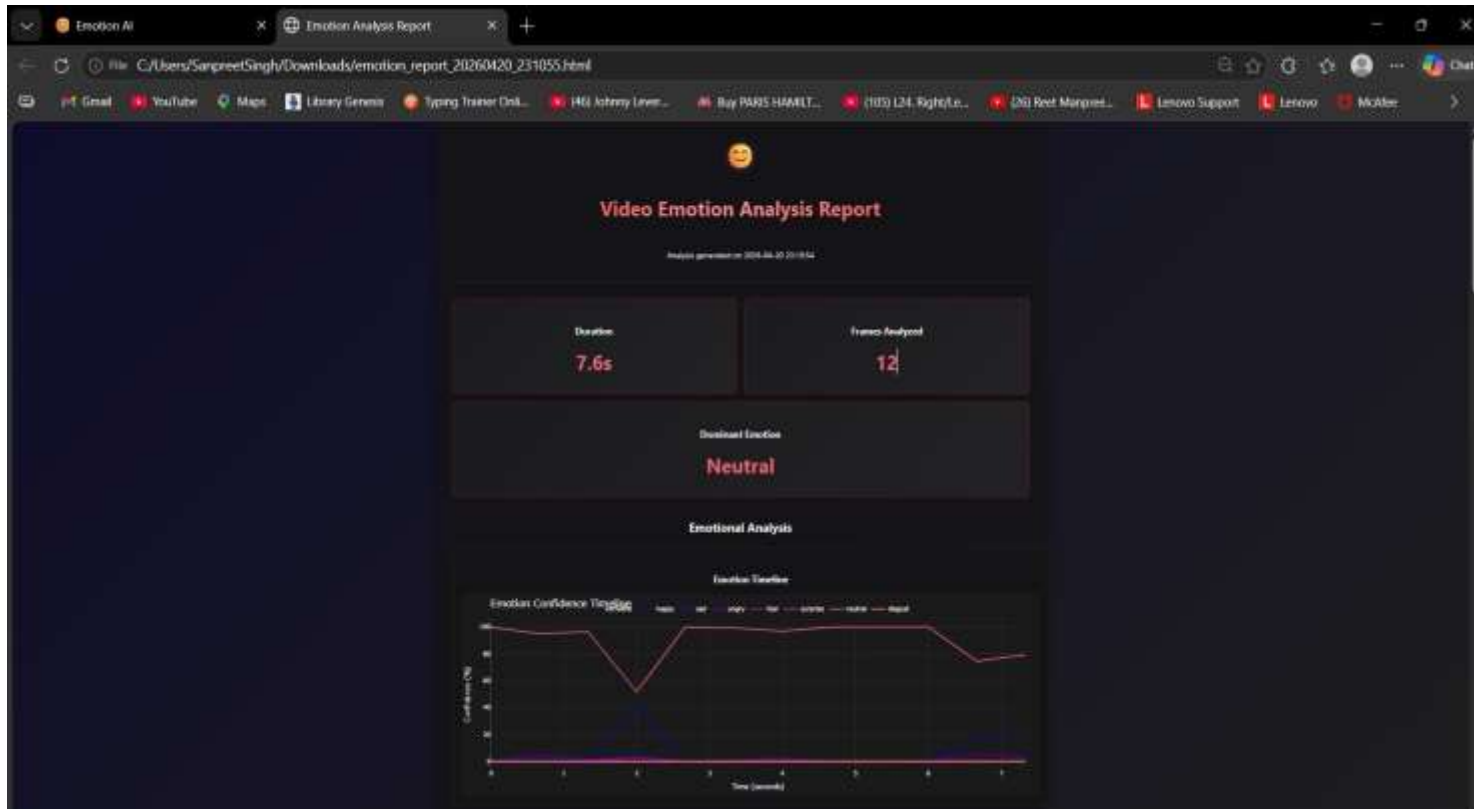


Report Generation:

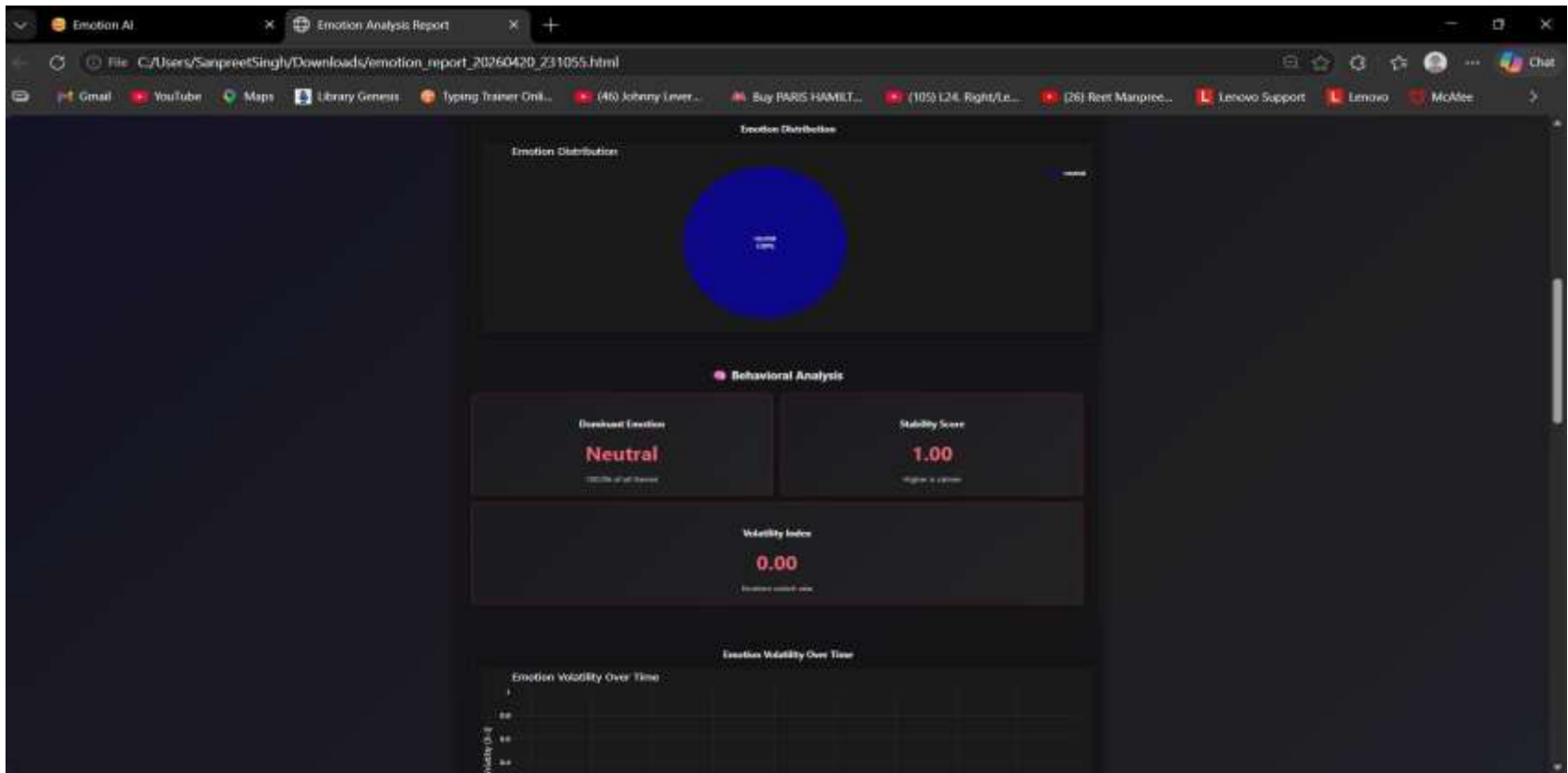
- Download HTML reports
- Charts + insights



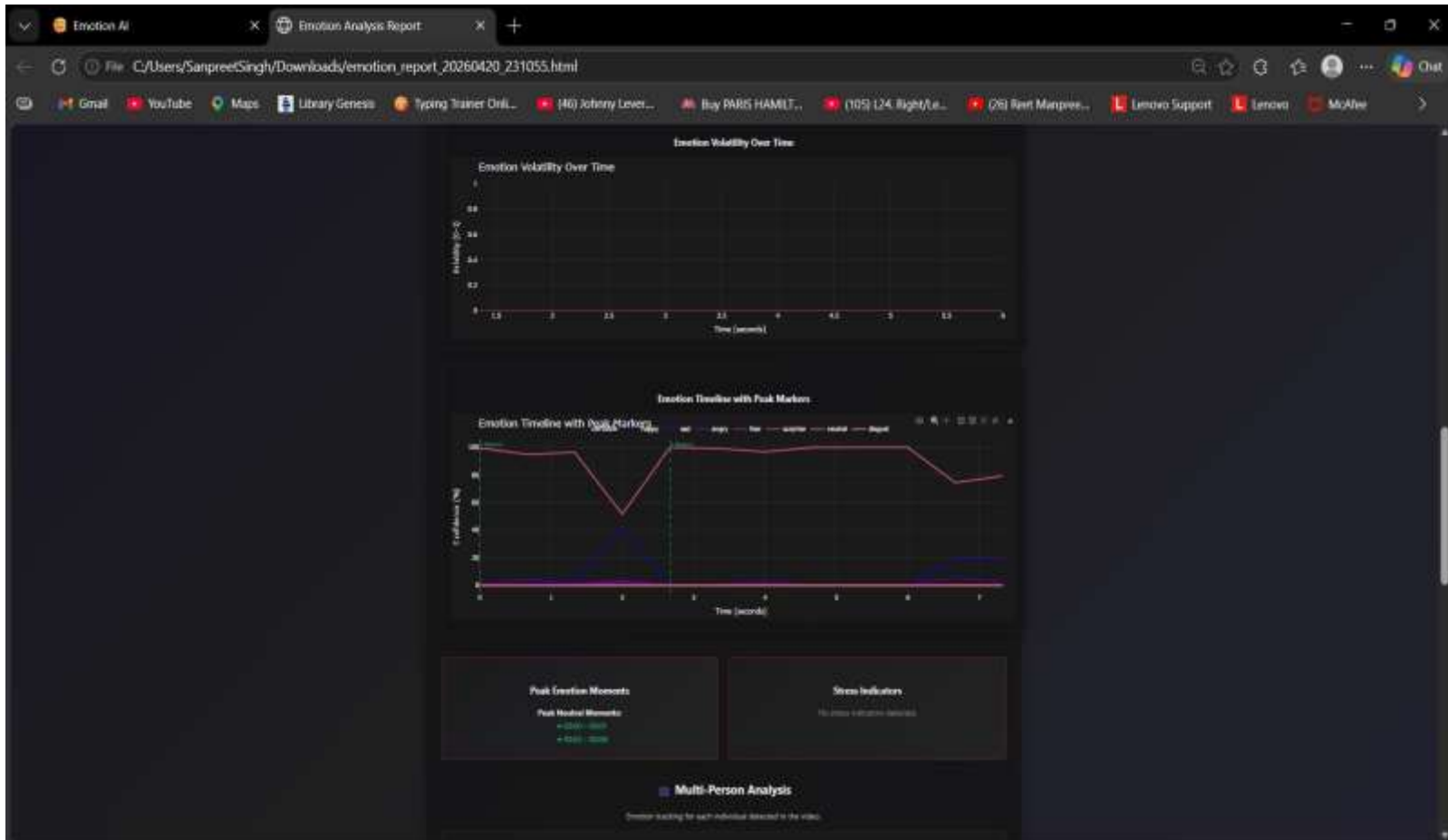
RESULTS



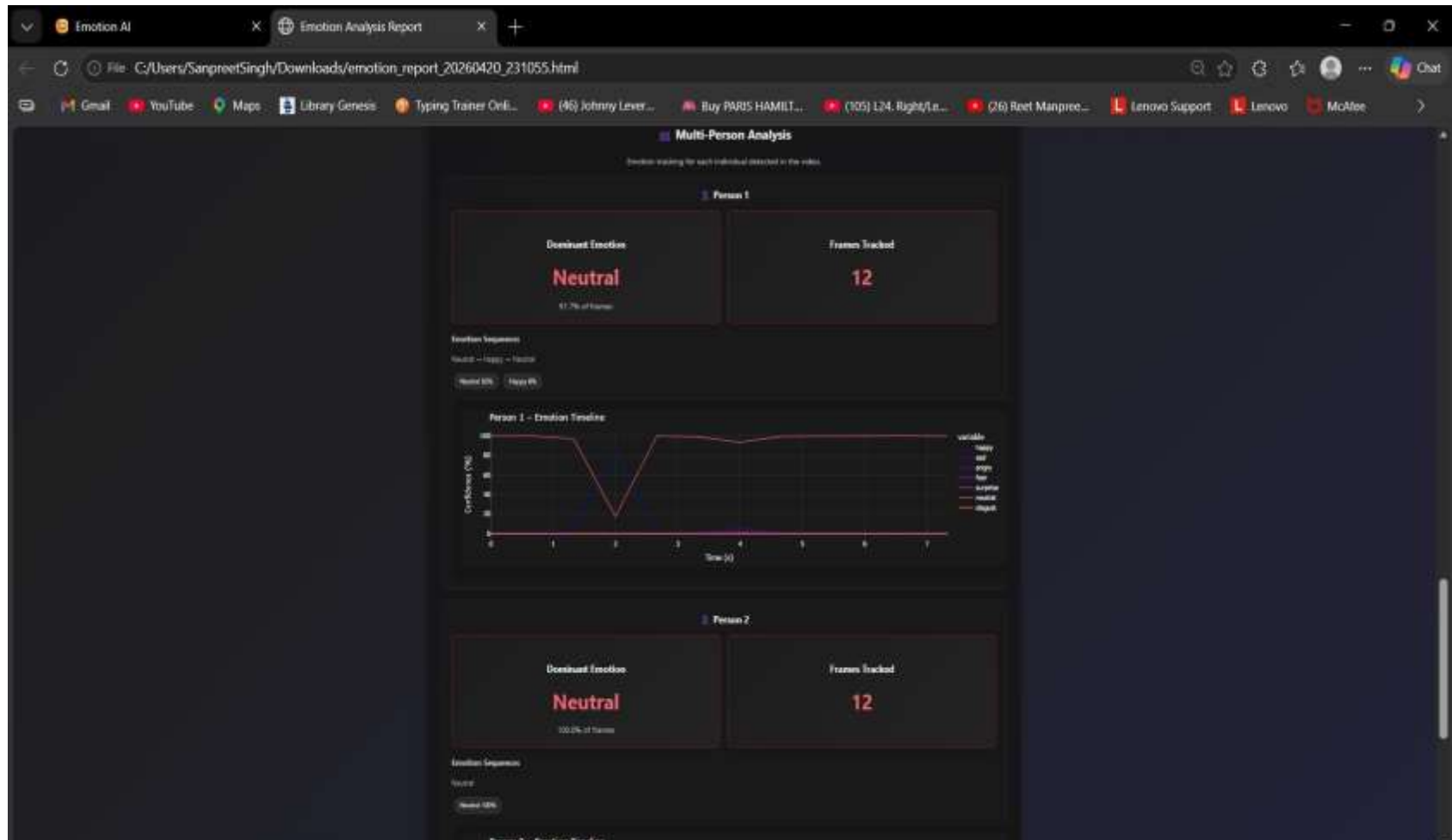
RESULTS



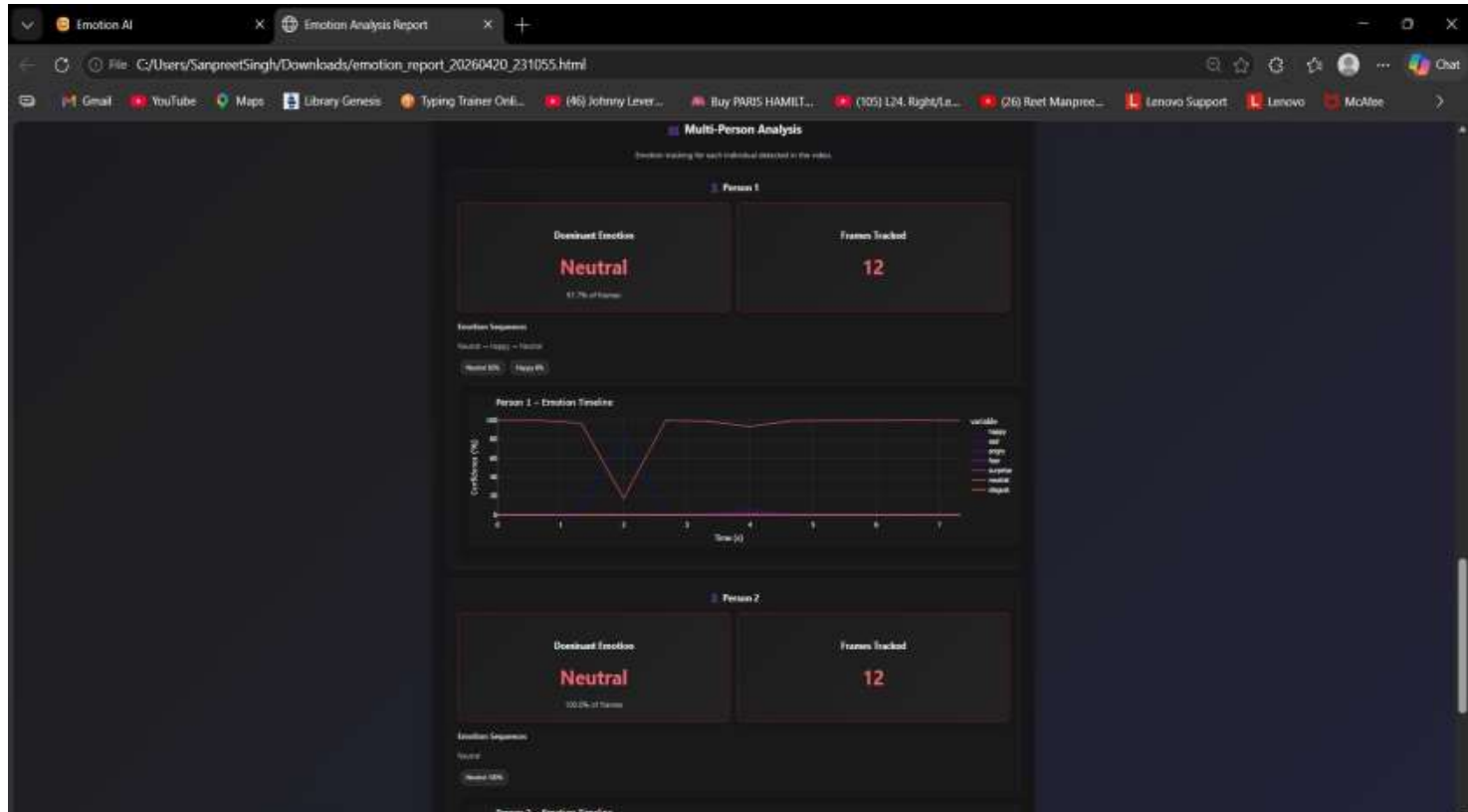
RESULTS



RESULTS



RESULTS



- **Advanced AI Models**

Integration of custom-trained deep learning models to improve accuracy in diverse conditions.

- **Multilingual Voice Support** 🌐

Extend voice emotion detection to support multiple languages and accents.

- **Cloud Deployment** ☁️

Deploy system on cloud for scalability, real-time access, and large-scale processing.

- **Mobile Application** 📱

Develop Android/iOS apps for on-the-go emotion detection and monitoring.

- **Wearable Integration** ⌚

Combine with smart devices (heart rate, stress sensors) for enhanced emotion analysis.

- **Enhanced Prediction Models** 🔮

Use advanced models like LSTM/RNN for better future emotion forecasting.

- **Real-Time Alert System** 📢

Generate alerts for stress, abnormal behavior, or emotional instability.

- **Privacy & Security** 🔒

Implement encryption and secure storage for sensitive emotional data.

The EmoGalaxy project successfully demonstrates the development of a multimodal emotion detection system that analyzes human emotions using facial expressions, voice signals, and behavioral patterns. The system is capable of performing both real-time and offline analysis through webcam, image, and video inputs. By integrating artificial intelligence, computer vision, and audio processing techniques, EmoGalaxy achieves more accurate and context-aware emotion recognition compared to traditional single-modality systems. Advanced features such as multimodal fusion, multi-person tracking, behavioral analysis, and emotion prediction further enhance the system's capabilities. The platform also provides interactive visualizations and detailed reports, making it useful for understanding emotional patterns effectively. Overall, EmoGalaxy highlights the potential of emotion-aware systems in real-world applications such as healthcare, surveillance, and human-computer interaction, and serves as a strong foundation for future advancements in this domain.

1. DeepFace Library Documentation

[DeepFace Documentation](#)

Official documentation for accessing photo used in the project.

2. OpenCV Official Documentation

[OpenCV Official Documentation](#)

Provides comprehensive resources and best practices for using OpenCV in computer vision, including face detection and image processing.

3. Scikit-learn: Machine Learning in Python

[Scikit-Learn Official Documentation](#)

Library used for implementing machine learning algorithms like TF-IDF and cosine similarity.

4. Cosine Similarity Explanation and Usage

[Cosine Similarity Official Documentation](#)

Wikipedia article explaining the concept and application of cosine similarity in text analysis.



***THANK
YOU***